

EXHIBIT 32

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON DC**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S SECOND SUPPLEMENTAL
OBJECTIONS AND RESPONSES TO RESPONDENTS' FIRST SET OF
INTERROGATORIES (NOS. 1-73)**

Pursuant to the United States International Trade Commission's Rules of Practice and Procedure, 19 C.F.R. §§ 210.27 and 210.29, Complainant GlobalFoundries U.S. Inc. ("GlobalFoundries" or "Complainant") hereby further respond to Respondents Tower Semiconductor Ltd.; Tower Partners Semiconductor, Ltd.; Tower Semiconductor Italy S.r.l.; Tower US Holdings, Inc.; Tower Semiconductor San Antonio, Inc.; Tower Semiconductor NPB Holdings, Inc.; Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC's (collectively "Tower" or "Respondents") First Set of Interrogatories Nos. 1-73.

GlobalFoundries's responses are based on information currently available to GlobalFoundries. GlobalFoundries reserves all rights to supplement, revise, and/or amend these responses, should additional information become available through the discovery process or by other means. In responding to these Interrogatories, GlobalFoundries does not waive any objection on the grounds of privilege, confidentiality, competency, relevance, materiality, authenticity, admissibility of the information contained in these responses, or any other objection.

INTERROGATORY NO. 28:

Provide all factual and legal bases for GlobalFoundries' contention that the Asserted Claims are valid under pre-AIA 35 U.S.C. § 102 and/or AIA 35 U.S.C. § 102, where applicable.

RESPONSE TO INTERROGATORY NO. 28:

GlobalFoundries incorporates all of its General Objections and reservations of rights as if set forth herein. GlobalFoundries objects to this Interrogatory as a premature contention interrogatory because Ground Rule 4.4.2 provides for contention interrogatory responses in accordance with the deadlines that will be set forth in the Procedural Schedule. GlobalFoundries further objects to this Interrogatory as improper burden-shifting because the invalidity burden rests with Respondents who have not yet provided any Invalidity Contentions. GlobalFoundries further objects to this Interrogatory to the extent it seeks information subject to attorney-client privilege, the work product doctrine, and any other applicable protections. GlobalFoundries further objects to this Interrogatory as overbroad and unduly burdensome because it seeks "all factual and legal bases" without reasonable limitation and is therefore disproportionate to the needs of this Investigation.

Subject to and without waiving the foregoing objections and the General Objections above, GlobalFoundries responds as follows: Pursuant to Ground Rule 4.4.2, GlobalFoundries will respond to contention interrogatories in accordance with the deadlines set forth in the Procedural Schedule.

FIRST SUPPLEMENTAL RESPONSE TO INTERROGATORY NO. 28 (July 30, 2026):

Subject to and without waiving the foregoing objections and the General Objections above, GlobalFoundries responds as follows:

GlobalFoundries responds to Tower's preliminary invalidity contentions. Tower bears the burden of proving invalidity by clear and convincing evidence and must establish that every

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limitation of every asserted claim is disclosed by or rendered obvious in view of the prior art. GlobalFoundries does not concede Tower's claim constructions, priority-date assertions, prior-art-status assertions, or characterizations of the cited references. This response is based on Tower's current contentions and the present record. GlobalFoundries preserves all validity arguments and reserves the right to respond to any amended, supplemented, or otherwise modified contentions.

I. U.S. Patent No. 8,330,235 ("235 Patent")

Tower has failed to meet its burden to establish by clear and convincing evidence that any prior art reference anticipates any asserted claim of the '235 patent. To anticipate, a single prior art reference must disclose each and every limitation of a claim, arranged as in the claim. Tower's five claim charts (Exhibits A1 through A5) each fail to establish that the charted reference discloses all limitations of the claims to which it is mapped.

Tower has failed to meet its burden to establish by clear and convincing evidence that Shimizu '954 discloses: a silicon substrate; a transistor comprising source/drain regions and a gate electrode on the substrate; a nickel silicide layer on the gate electrode and source/drain regions, wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode, from between about 8 wt.% and about 12 wt.% at the silicon interface; a nickel silicide layer having a thickness of about 100 Å to about 150 Å; a silicon substrate comprising a strained silicon germanium (Si-Ge) layer on a silicon layer; a stress liner on the nickel silicide layer; an oxide or oxynitride layer on the stress liner; a stress liner comprising stressed silicon nitride; a P-channel transistor and an N-channel transistor, each comprising source/drain regions

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testimony that exceeds the scope of Tower's preliminary contentions.

II. U.S. Patent No. 8,507,983 ("’983 Patent")

Tower has failed to meet its burden to establish by clear and convincing evidence that any prior art reference anticipates any asserted claim of the ’983 patent. To anticipate, a single prior art reference must disclose each and every limitation of a claim, arranged as in the claim. Tower's thirteen claim charts (Exhibits B1 through B13) each fail to establish that the charted reference discloses all limitations of the claims to which it is mapped.

Tower has failed to meet its burden to establish by clear and convincing evidence that Shrivastava ’773 discloses: a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric; a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; a doped drift well adjacent to a second edge of the gate; a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region; a drift isolation region in the doped drift well, and a doped deep well in the substrate of the active device region encompassing the doped drift well; the drift isolation region disposed under the gate; the first and a second heavily doped regions provided adjacent to the gate, wherein the first heavily doped region is disposed in the doped channel well on a source side of the device, and the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device; the first edge of the gate corresponding to a channel edge of the gate of the device and a second edge of the gate corresponding to a drain edge of the gate of the device; or the



displacement of the channel Lch from the first heavily doped region.

For example, for the asserted “drift isolation region... disposed under the gate” limitation, Tower switches embodiments and maps to the “thick field oxide 12” of the Fig. 1a embodiment—a different, non-self-aligned embodiment that Chen ’375 itself criticizes: “the drain region 37 is separated from the edge of gate structure 36 with a space S. When misalignment occurs... an unstable conductive resistance RON... reduce[s] process stability.” Chen ’375 at 2:1-26; Ex. B3 at 6-7. Tower has failed to show how that embodiment includes a self-aligned sidewall-spacer/ion-implantation structure that Tower relies upon from the Fig. 3a-3e embodiment for the channel-well, drift-well, spacer, and channel-length limitations.

Tower has failed to meet its burden to establish by clear and convincing evidence that Wu ’291 discloses: a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric; a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; a doped drift well adjacent to a second edge of the gate; a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region; a drift isolation region in the doped drift well, and a doped deep well in the substrate of the active device region encompassing the doped drift well; the drift isolation region disposed under the gate; the first and a second heavily doped regions provided adjacent to the gate, wherein the first heavily doped region is disposed in the doped channel well on a source side of the device, and the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the



device; the first edge of the gate corresponding to a channel edge of the gate of the device and a second edge of the gate corresponding to a drain edge of the gate of the device; or the channel having a channel length that is controlled by ion implantation.

For example, Tower has failed to identify and explain how “the first edge of the gate and channel edge beneath the gate define a channel” or how Wu ’291 discloses the claimed channel. For example, Tower has failed to show that any channel defines the displacement of the unspecified channel from the first heavily doped region, including because Tower has failed to identify any alleged displacement in Wu ’291. For example, Tower has failed to identify any device isolation region as claimed. For example, Tower has failed to identify that the second edge of the gate corresponds to the drain edge of the gate. For example, Tower fails to explain how the element 210 is a “doped channel well” as claimed. For example, Tower has not identified a cited passage in Wu ’291 establishing that element 210 (the mapped “doped drift well”) is positioned “adjacent to a second edge of the gate” as the asserted limitation requires. Wu ’291’s disclosed structure shows element 116 (field oxide) between the gate edge and the drain region, and Tower has not established the asserted adjacency relationship between the drift well and the gate’s second edge from Wu ’291’s cited passages.

For example, Wu ’291 states that “the gate dielectric 310 and the gate electrode 312 are formed over the interposed region 214 such that the interposed region 214 forms a portion of the channel region of the transistor,” and describes interposed region 214 as “preferably a portion of the substrate 110 above the deep N-well 112” having a dopant concentration of “about $1\text{E}14$ to about $1\text{E}17$ atoms/cm³.” Wu ’291 at [0036], [0039]. Wu ’291 further specifies that “the P-well 210 and the N-well 212 are not immediately adjacent to each other,” and that the distance between them “is preferably between about 0.2 μm and 1.5 μm .” *Id.* at [0036]. Tower has not identified a

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cited passage establishing that an edge of P-well 210 beneath the gate, together with the first edge of the gate, defines the channel of Wu '291's transistor, as the asserted limitation requires. Tower's mapping does not establish the asserted channel-well limitation.

For example, as to the asserted doped-deep-well limitation, Tower relies on Wu '291's deep N-well 112. Wu '291 describes deep N-well 112 as formed "in the high-voltage PMOS region 104" and as preferably "formed about 1 um to about 6 um below the surface of the substrate 110, thereby leaving a portion of the substrate 110 between the deep N-well 112 and the surface of the substrate 110." Wu '291 at [0029]. Wu '291's own claims describe that element as "a deep well subjacent to the first well or the second well" and as "the third well being immediately below the first well or the second well." Wu '291, claims 1, 13. Tower has not identified a cited passage establishing that a well positioned subjacent to, or immediately below, the mapped drift well "encompass[es]" that drift well as the asserted limitation requires. Tower has also not reconciled its reliance on Wu '291's high-voltage NMOS structure for other asserted limitations with Wu '291's description of deep N-well 112 as formed in the high-voltage PMOS region.

Tower has failed to meet its burden to establish by clear and convincing evidence that Kuzuhara '978 discloses: a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric; a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; a doped drift well adjacent to a second edge of the gate; a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region; a drift isolation region



device; the first edge of the gate corresponding to a channel edge of the gate of the device and a second edge of the gate corresponding to a drain edge of the gate of the device; or the channel having a channel length that is controlled by ion implantation.

For example, Tower's identification of "the left edge of the gate to the right edge of element 104" is nonsensical and fails to disclose the claimed displacement in any event. For example Tower maps the asserted "doped channel well" to element 104 (p-well) using the Fig. 8 alternative embodiment where "p-well 104 has been laterally aligned to an edge of first gate electrode 121 so that p-well 104 is substantially confined to a region under source/drain region 145." Yang '928 at [0045]; Ex. B10 at 3-4. Tower maps the asserted "doped drift well" to element 106. For the asserted "doped deep well" limitation, Tower's chart maps to "element 107 or element 103" in the disjunctive. Ex. B10 at 7. Tower has not committed to which of two different disclosed elements—a well region versus the epitaxial layer itself—satisfies the claimed "doped deep well... encompassing the doped drift well," leaving the mapping ambiguous. A disjunctive mapping that identifies two different structural elements as alternative candidates does not constitute clear and convincing evidence. For example, Tower has failed to explain how element 104 is the claimed doped channel well. For example, Tower has failed to explain how the first edge of the gate overlaps the channel well 104, or how the first edge of the gate and channel edge beneath the gate define a channel. For example, Tower has failed to identify a doped drift well adjacent to the second edge of the gate and has not identified any spacer that defines the alleged displacement.

For example, Tower relies on the Fig. 8 alternative embodiment of Yang '928, but Tower has not established that this embodiment discloses all asserted limitations in combination. Tower has not identified a cited passage establishing that the Fig. 8 embodiment includes both the asserted "drift isolation region in the doped drift well" and the asserted "doped deep well... encompassing

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the doped drift well” simultaneously with the laterally-aligned p-well arrangement Tower relies upon for the channel-well limitation.

Tower has failed to meet its burden to establish by clear and convincing evidence that Voldman '811 discloses: a substrate prepared with an active device region, wherein the active device region includes a gate, the gate including a gate electrode over a gate dielectric; a doped channel well disposed in the substrate adjacent to a first edge of the gate, the first edge of the gate overlaps the channel well with a channel edge of the channel well beneath the gate, the first edge of the gate and channel edge beneath the gate define a channel of the device, wherein the channel is displaced from a first heavily doped region adjacent to the first edge of the gate; a doped drift well adjacent to a second edge of the gate; a spacer on the first edge of the gate, wherein the spacer defines the displacement of the channel from the first heavily doped region; a drift isolation region in the doped drift well, and a doped deep well in the substrate of the active device region encompassing the doped drift well; the drift isolation region disposed under the gate; the first and a second heavily doped regions provided adjacent to the gate, wherein the first heavily doped region is disposed in the doped channel well on a source side of the device, and the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device; the first edge of the gate corresponding to a channel edge of the gate of the device and a second edge of the gate corresponding to a drain edge of the gate of the device; or the channel having a channel length that is controlled by ion implantation.

For example, Tower has failed to identify the claimed displacement, or any spacer that defines the alleged displacement. For example, Tower has failed to explain how element 424 comprises the claimed doped drift well adjacent to a second edge of the gate. Tower maps the asserted “drift isolation region” to element 429 and the asserted “doped drift well” to element 424,



drawing on a disclosed “guard ring” structure (420'+412'+424'+428') that Voldman '811 describes as formed “simultaneously” with, and structurally parallel to, an LDMOS transistor 470. Ex. B11 at 5-6. Tower has not established whether its cited passages describe the claimed transistor's own doped-well architecture or instead describe a separate guard-ring structure whose fabrication merely parallels the transistor's. Tower has not shown that the asserted limitations are satisfied by the LDMOS transistor structure itself, as opposed to a functionally distinct guard-ring element.

For example, Tower's mapping draws on Voldman '811's Fig. 4, but Voldman '811's description of that figure states only that “the LDMOS transistor 470 is formed by a conventional method,” and otherwise describes the formation of the primed guard-ring regions 420', 412', 424', and 428'. Voldman '811 at [0038]–[0039]. Tower has not identified a cited passage describing the structural or doping relationships among the unprimed transistor regions on which its mapping depends. For example, Voldman '811's claims are directed to a guard ring that “forms a closed loop around the first transistor” and to methods of simultaneously forming transistor and guard-ring regions, Voldman '811, claims 1, 9, and Tower has not identified any Voldman '811 claim reciting the asserted channel well, drift well, drift isolation region, or spacer-defined displacement. For example, in the Fig. 1G sequence that Voldman '811 does describe in structural detail, the P-body region 122 is formed by implantation into N-region 120 before the STI regions 124 are formed and before the gate dielectric region 130 and gate electrode region 140 are formed, and the extension region 123 is formed before spacer regions 150. Voldman '811 at [0019]–[0024]. Tower has not identified a cited passage establishing that the resulting boundary of the mapped channel well beneath the gate defines a channel together with the first edge of the gate, or that spacer regions 150—which Voldman '811 states “can be formed by a conventional method,” *id.* at

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from each other by a channel expected distance.” Lee ’935 at [0008]-[0009]; see also *id.*, claims 1, 2, 16, 19. Lee ’935 further explains that “the entire channel region C is formed in the p body region 160,” with the stated objective that “the channel region C has a uniform impurity profile.” *Id.* at [0026]. Tower has not identified a cited passage establishing that the channel of Lee ’935’s device is bounded by the first edge of the gate and an edge of a doped channel well beneath the gate, as opposed to by the spacing between the two channel stop regions that Lee ’935 itself identifies. Tower has likewise not shown that the mapped p body region 160 is a “doped channel well disposed in the substrate adjacent to a first edge of the gate” having “a channel edge of the channel well beneath the gate,” where Lee ’935 describes that body region as underlying both channel stop regions and both gate edges.

For example, Lee ’935 identifies the resulting device as “a both sides diffused MOS (BDMOS) transistor,” in which “the first and second channel stop regions 170a and 170b ate [sic] formed and diffused on both sides of the p body region 160.” Lee ’935 at [0025]. Tower has not explained how a structure in which two like-doped channel stop regions are symmetrically disposed on both sides of a single body region discloses the asserted asymmetric arrangement of a doped channel well adjacent to the gate’s first edge and a doped drift well adjacent to the gate’s second edge. Tower’s mapping does not establish the asserted limitations.

For example, for the asserted drift-isolation-region limitations, Tower relies on Lee ’935’s local oxide layer 175. Lee ’935 states that “[t]he local oxide layer 175 can serve as a gate insulating layer for the power device and, therefore, is a thick film,” describes it as formed “in a predetermined portion of the surface of the semiconductor substrate,” and describes the gate electrode 190 as subsequently formed “over part of the first channel stop region 170a, part of the p body region 160, part of the second channel stop region 170b, and the local oxide layer 175.”

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Lee '935 at [0022]–[0023]. Tower has not identified a cited passage establishing that an element Lee '935 characterizes as a gate insulating layer is a “drift isolation region in the doped drift well.” For example, as to the asserted doped-deep-well limitation, Lee '935 describes an n⁺ burying layer 110 formed “deep in a semiconductor substrate 100,” with the n[–] well 120 formed “above the n⁺ burying layer 110,” Lee '935 at [0018], and Tower has not identified a cited passage establishing that any Lee '935 element encompasses the mapped drift well. For example, as to the asserted arrangement in which the second heavily doped region is disposed between the drift isolation region and a device isolation region on the drain side of the device, Tower has not identified any Lee '935 element as the claimed device isolation region.

For example, Lee '935 describes spacer 195 as “formed by any well-known method on both sidewalls of the gate electrode 190,” after which “N⁺ impurities are then ion-implanted into the first and second channel stop regions 170a and 170b on both sides of the gate electrode 190 to form a source region 220a and a drain region 220b.” Lee '935 at [0024]. Tower has not identified a cited passage establishing that spacer 195 defines any displacement of the channel from the first heavily doped region, as the asserted limitation requires. At a minimum, Tower has failed to establish the asserted spacer-defined displacement by clear and convincing evidence.

Tower has separately failed to meet its burden to establish by clear and convincing evidence that any charted reference discloses the asserted limitation that the channel have a channel length that is controlled by ion implantation. Tower's charts recite this limitation but do not identify, for each charted reference, a cited passage establishing that the length of the channel—as distinct from the formation of some doped region by implantation—is controlled by an implantation parameter. For example, Lee '935 describes the channel as formed “between the first and second channel stop regions,” which “can be spaced apart from each other by a channel

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expected distance,” and describes the body region and the channel stop regions as formed by separate masked implants performed at comparable energies “of about 90 to about 110 KeV.” Lee ’935 at [0009], [0019]-[0020]. For example, Wu ’291 describes the region that “forms a portion of the channel region of the transistor” as the interposed region 214, and states that “[t]he distance between the P-well 210 and the N-well 212 is preferably between about 0.2 um and 1.5 um,” with a gate length “preferably about 1 um to about 3 um.” Wu ’291 at [0036], [0039]. For example, in Shrivastava ’773’s cited process flow, first well 702 and second well 704 are implanted before the gate region 722 is formed, and Tower has not identified a cited passage establishing that any implantation parameter controls the length of the resulting channel. Shrivastava ’773 at [0063]-[0065].

Tower has also failed to meet its burden to establish by clear and convincing evidence that any charted reference discloses the asserted limitations that the first edge of the gate correspond to a channel edge of the gate of the device and the second edge of the gate correspond to a drain edge of the gate of the device. Tower’s charts identify gate edges by their orientation within a selected figure—for example, “the left edge of the gate electrode 3” or “the left edge of the gate to the right edge of element 104”—without identifying a cited passage establishing which edge of the gate the reference itself associates with the channel of the device and which with the drain. Ex. B5 at 2-5; Ex. B10 at 3-4. A figure-orientation identification does not establish the asserted correspondence between gate edges and the channel and drain edges of the device.

Tower has further failed to meet its burden to establish by clear and convincing evidence that any charted reference discloses the asserted arrangement in which the second heavily doped region is disposed between the drift isolation region and a device isolation region on a drain side of the device. That limitation recites an ordering of three separately identified elements, and for

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several charted references Tower identifies isolation structures only generically. For example, Levin '977 describes isolation oxide structures 111, 112, 115, 117, and 118 as formed “using known STI processes” or, “[i]n alternative embodiments, ... using poly buffered local oxidation of silicon (PBLOCOS) or standard LOCOS techniques,” and locates the N+ drain implant 193B “between central isolation region 115 and peripheral isolation region 112.” Levin '977 at 3:52-58, 4:50-56. For example, Voldman '811 describes STI regions 124 as formed by patterning a photoresist layer, anisotropically etching shallow trenches, and “filling back the shallow trenches with silicon dioxide.” Voldman '811 at [0020].

Tower has likewise failed to meet its burden to establish by clear and convincing evidence the asserted limitation of a doped deep well encompassing the doped drift well with respect to additional charted references. For example, Tower relies on Morton '874's statement that “[t]o provide isolation of the p-well drain extension region 70 from the p-type substrate 10, a deep n-well region 150 is formed by implanting phosphorous or arsenic into the substrate,” and that “[t]he deep n-well region 150 serves to isolate the p-well region 70 from the substrate 10.” Morton '874 at 4:62-65, 5:31-34; Tower Contentions at 171–172. For example, Tower relies on You '507's statement that the disclosed process includes “implanting n-type dopants into a surface of the substrate to form an n-doped deep n-well (DNW) region.” You '507 at 1:57-59; Tower Contentions at 173. For example, Tower relies on Klein 2008's statement that “[t]he device breakdown voltage is initially set by the dose of the high voltage Nwell implant conditions.” Klein 2008, p. 2; Tower Preliminary Invalidity Contentions at 174. Tower has failed to establish that any references disclose the claimed doped deep well or explained how its cited excerpts purportedly disclose the claimed doped deep well.

Tower has also failed to meet its burden to establish by clear and convincing evidence the

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asserted limitation that the drift isolation region be disposed under the gate with respect to additional charted references. For example, Tower relies on Klein 2008's statement that "[t]he poly overlap on STI enhances the depletion of the drain when the device is in the off state." Klein 2008, p. 2; Tower Preliminary Invalidity Contentions at 139-141. Tower has not identified a cited passage establishing that the STI structure described in that passage is disposed in a doped drift well, as the asserted limitations separately require, or that the described overlap corresponds to the asserted under-the-gate arrangement. For example, Tower relies on You '507's statement that "a transistor gate at least partially covers a regular gate oxide plus a drain oxide (e.g., such as formed through LOCOS)." You '507 at 6:44-57; Tower Preliminary Invalidity Contentions at 147-148. Tower has not identified a cited passage establishing that the drain oxide so described is disposed in the mapped doped drift well.

Tower has failed to meet its burden to establish by clear and convincing evidence that the asserted claims are obvious. Tower purports to identify various disparate references allegedly disclosing certain claimed features. But Tower has not shown that any specific combination of references renders any specific claim of the '983 patent obvious.

Tower's obviousness contentions are structured as element-by-element "groupings" of references—collections of disparate prior art references that Tower asserts, individually, disclose a particular claim limitation. Tower Preliminary Invalidity Contentions at 117-174. But grouping references by feature is not an obviousness analysis. Tower never explains how any specific combination of references would be made; Tower never identifies the particular combination it will rely upon for any given claim; Tower never explains how any combination would result in an operable high-voltage LDMOS device; and Tower never provides the required particularized motivation to combine any specific set of references. Tower's approach reduces to the

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impermissible assertion that each feature was independently “standard” or “commonly found” in MOSFET devices—a hindsight-driven “obvious to try everything” framework that cannot substitute for the particularized motivation-to-combine showing required by law.

Tower’s “Exemplary Obviousness Combinations” section compounds these deficiencies. Rather than identifying a specific combination and explaining why a POSITA would have been motivated to combine those specific references with a reasonable expectation of success, Tower presents undifferentiated “and/or” permutation lists that generate an extremely large number of unspecified combinations. *See, e.g.*, Tower Preliminary Invalidity Contentions at 89. These formulations generates hundreds of permutations for a single primary reference—and Tower provides similar lists for each of its primary references, resulting in thousands of theoretical combinations for the ’983 patent alone. Such shotgun arguments provide no notice of any actual theory and does not satisfy Tower’s burden. For the ’983 patent specifically, a POSITA seeking to combine teachings from disparate LDMOS/high-voltage transistor references would face substantial integration challenges that Tower has not addressed.

Tower’s Shrivastava ’773 permutation list alone illustrates the deficiency. That single formulation offers six alternative references for one asserted element, six for another, and seven for a third, each set joined to the others by “and/or,” and each set also available in the alternative to Shrivastava ’773 “alone.” Tower Preliminary Invalidity Contentions at 89. Taken at face value, that one list describes 392 distinct combinations, and Tower does not identify which of them it intends to advance against any asserted claim. Tower provides comparable “and/or” lists for each of its other twelve primary references. Tower Preliminary Invalidity Contentions at 88-117. Tower has not identified any specific combination, for any specific asserted claim, together with a reason a person of ordinary skill in the art would have made that combination and a basis on

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the work product doctrine, and any other applicable protections. GlobalFoundries further objects to this Interrogatory as vague and ambiguous because the phrases “policy or policies,” “monitor,” and “patent-related filings” are undefined and susceptible to multiple interpretations.

In light of the foregoing objections, GlobalFoundries will not provide a response to Interrogatory No. 72.

INTERROGATORY NO. 73:

Describe all facts and circumstances related to your pre-suit investigation and preparation of the Complaint in This Investigation.

RESPONSE TO INTERROGATORY NO. 73:

GlobalFoundries incorporates all of its General Objections and reservations of rights as if set forth herein. GlobalFoundries objects to this Interrogatory to the extent it seeks information protected by the attorney-client privilege or work-product doctrine. GlobalFoundries further objects to this Interrogatory as vague and ambiguous because the phrases “facts and circumstances,” “pre-suit investigation,” and “preparation” are undefined and susceptible to multiple interpretations. GlobalFoundries further objects to this Interrogatory to the extent it seeks information that is irrelevant to the claims or defenses in this Investigation. GlobalFoundries further objects to this Interrogatory as overbroad and unduly burdensome because it seeks “all facts and circumstances” without reasonable limitation and is therefore disproportionate to the needs of this Investigation.

In light of the foregoing objections, GlobalFoundries will not provide a response to Interrogatory No. 73.

Dated: July 30, 2026

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CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S SECOND SUPPLEMENTAL OBJECTIONS AND RESPONSES TO RESPONDENTS' FIRST SET OF INTERROGATORIES (NOS. 1-73)** were served on July 30, 2026 as follows:

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/s/ Erika J. Weinstein

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